

CubeXHome

Comprehensive Property Security & Management Platform

Status: In Development • Platform: Web & Mobile • Market: Malaysia

CubeXHome is a comprehensive property security and management platform designed specifically for Malaysian condominiums, apartments, and gated communities. The platform connects all stakeholders—residents, security guards, property managers, and administrators—into a unified ecosystem that prioritizes security, convenience, and operational efficiency.

The system addresses real-world challenges faced by property management companies, security firms, and residents by digitizing and automating traditionally manual processes while maintaining enterprise-grade security standards.

The Problem

Property management in Malaysia faces several persistent challenges that affect all stakeholders in the ecosystem:

For Residents

- No real-time visibility into security operations or guard patrol status
- Manual visitor registration processes that are time-consuming
- Delayed communication from property management
- Lack of emergency alert system for urgent situations
- Paper-based payment and feedback submission systems

For Security Personnel

- Paper-based patrol logging that is prone to falsification
- No accountability system for checkpoint completion verification
- Delayed incident reporting and documentation
- No centralized emergency coordination system

For Property Managers

- Manual billing and payment tracking processes
- No real-time oversight of security operations
- Fragmented communication channels across different platforms
- Difficulty tracking and resolving maintenance requests

For Property Owners & Developers

- No visibility across multiple properties under management
- Manual subscription and payment collection from properties
- Lack of standardized operations across different sites
- Difficulty enforcing consistent service quality standards

The Solution

CubeXHome provides a multi-platform ecosystem that digitizes every aspect of property security and management. The platform consists of four integrated components, each designed for specific user roles while sharing a unified backend infrastructure.

Platform	Users	Access Type
CubeXHome Admin	Super Administrators	Web Dashboard
Site Management	Property Managers	Web Dashboard
Security Dashboard	Security Heads	Web Dashboard
Mobile Application	Residents & Guards	iOS & Android

Key Features by User Role

Residents (Mobile App)

- **Digital ID (CubeX ID)** — QR-based identification for visitor verification and access control
- **SOS Emergency** — One-tap panic button with automatic location sharing to security
- **Visitor Management** — Pre-register visitors and receive real-time arrival notifications
- **Bill Payments** — View outstanding fees and make payments directly through the app
- **Announcements** — Receive real-time updates and notices from property management
- **Feedback System** — Submit and track maintenance requests and complaints

Security Guards (Mobile App)

- **Biometric Clock-In** — GPS-verified attendance with biometric authentication
- **Digital Patrol System** — QR/NFC checkpoint scanning with photo evidence capture
- **Incident Reporting** — Real-time documentation with media attachments
- **Panic Button** — Emergency alert broadcast to all nearby guards and supervisors
- **Visitor Verification** — Scan and verify visitor QR codes at entry points

Security Head (Web Dashboard)

- **Live Guard Tracking** — Real-time location and status monitoring of all guards
- **Patrol Monitoring** — Track checkpoint completion rates and identify coverage gaps
- **Incident Management** — Review, escalate, and resolve reported incidents

- **Emergency Coordination** — Centralized emergency response management system

Site Manager (Web Dashboard)

- **Resident Management** — Complete resident profiles, unit assignments, and access control
- **Billing & Invoicing** — Generate bills, track payments, and send automated reminders
- **Feedback Management** — Track and resolve resident complaints with status updates
- **Announcements** — Broadcast messages to all residents via push notifications

Security Architecture

Security is the cornerstone of CubeXHome. The platform implements multiple layers of protection to ensure data integrity, user privacy, and operational safety.

Multi-Layer Authentication System

- **Layer 1 — Identity Verification:** Email and password authentication for web dashboards, email-based OTP verification for all users on new devices
- **Layer 2 — Device Trust:** Device fingerprinting and registration system, new device access triggers additional verification steps, users can view and revoke trusted devices
- **Layer 3 — Two-Factor Authentication:** TOTP-based 2FA compatible with Google Authenticator, mandatory for Super Admin accounts, optional for other roles
- **Layer 4 — Biometric Security:** Face ID and Touch ID support for mobile app, biometric data stored locally on device for privacy, PIN code available as backup method
- **Layer 5 — Session Management:** Role-based session duration limits, automatic timeout for sensitive operations, remote session revocation capability

Security Features

Feature	Implementation
Data Encryption	AES-256 encryption for all sensitive data at rest
Transport Security	TLS 1.3 protocol for all API communications
Token Security	JWT tokens with short expiry and refresh token rotation
Rate Limiting	Protection against brute force and DDoS attacks
Audit Logging	Complete trail of all security-related events
SQL Injection Prevention	Parameterized queries via Prisma ORM

Service Killswitch System

A unique automated payment enforcement feature for property service providers:

- **7 Days Before Due:** First automated reminder sent via email
- **3 Days Before Due:** Second automated reminder sent via email
- **On Due Date:** Final warning notification sent
- **7 Days After Due:** Services automatically suspended for the property

When suspended, all services are disabled including mobile app access, dashboard access, and notifications. When payment is received, services are restored immediately with all data preserved.

Technology Stack

Backend Infrastructure

Technology	Purpose
Node.js + Express	API server and runtime environment
PostgreSQL	Primary relational database
Prisma ORM	Type-safe database operations
JWT	Secure authentication tokens
Resend	Transactional email delivery
Firebase Cloud Messaging	Cross-platform push notifications

Frontend Technologies

Platform	Technologies Used
Web Dashboards	React 18, TypeScript, Vite, TailwindCSS, Framer Motion
Mobile Application	React Native, Expo, SecureStore, LocalAuthentication

Design System

Each platform component features a distinct visual identity while maintaining brand consistency:

Platform	Primary Color	Design Theme
Mobile App	Forest Green (#065f46)	Fresh, Trustworthy
Security Dashboard	Teal/Cyan (#0891b2)	Alert, Professional
Site Management	Purple (#7c3aed)	Premium, Sophisticated
Super Admin	Crimson (#dc143c)	Powerful, Authoritative

Project Status

Completed Components

- ✓ Backend API — All endpoints implemented and tested (100%)
- ✓ Database Schema — All tables and relationships defined (100%)
- ✓ Mobile App UI — All screens for Resident and Guard roles (100%)
- ✓ Security Dashboard UI — Complete with 8 functional pages (100%)
- ✓ Site Management UI — Complete with 9 functional pages (100%)
- ✓ Super Admin UI — Complete with killswitch system (100%)

In Progress

- API Integration — Connecting all frontend applications to backend
- Authentication System — Implementing OTP, 2FA, and biometric features
- Push Notifications — Setting up Firebase Cloud Messaging
- Testing & QA — Security audit and quality assurance

Expected Impact

Stakeholder	Expected Benefits
Property Management	40% reduction in administrative work, automated billing and reminders
Security Firms	Elimination of falsified patrols, faster incident response times
Residents	Enhanced safety with one-tap emergency alerts, digital convenience
Property Owners	Real-time visibility and control across all managed properties

Conclusion

CubeXHome represents a comprehensive solution to the fragmented property management landscape in Malaysia. By unifying all stakeholders on a single, secure platform, it addresses the core challenges of security accountability, operational efficiency, and resident satisfaction.

The emphasis on security—from multi-layer authentication to encrypted communications—ensures that sensitive resident data and property operations are protected to enterprise standards. As development continues, CubeXHome is positioned to become the standard for smart property management in the Malaysian market.

This case study represents a project currently in development.
Features and specifications are subject to change.

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